

TRAJECTORY GENERATION METHODOLOGY

Objective:

Our Objective is to analyse and compare the trajectories obtained for the same walking cycle using the same conditions when we use a Cubic curve, Half-Sine curve and a Quintic Curve.

Walking Cycle:

Initially the first walking cycle will consist of a starting phase in which the robot's leg will position itself to start walking, after that there will be a swing and a stance phase. After the first cycle each cycle will consist of two phases that is swing and stance.

1-2 : Starting Phase

2-3-4: Swing Phase

4-1-2: Stance Phase

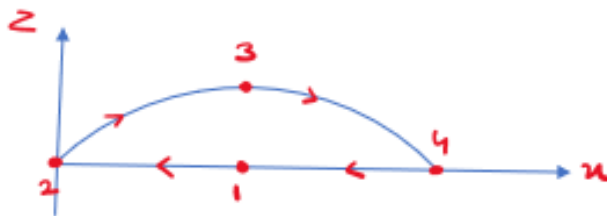


Figure: Walking Cycle Representation

Assumptions:

We have assumed that the time duration of the swing phase and the stance phase will be equal i.e. half of the total duration of the walking cycle's time period.

Let T_y be swing time period and T_r be the stance time period

$$T_y = T_r = T/2$$

We calculate each part of the problem from $t=[0, T_{\text{duration}}]$

Choice of Curves:

To satisfy the boundary conditions and get the desired shape of trajectory in space the curves to be used are cubic curve, half sine curve and quintic curve.

Input Values:

The following values are asked for as input

T: Total time of one walking cycle

S_0 : Length of the footstep

H_0 : Height of step

To verify our results the following sample inputs are used throughout

T: 0.7 s

S_0 : 450 mm

H_0 : 100 mm

Methodology:

To analyse and compare the trajectories obtained for the same walking cycle using the same conditions when we use a Cubic curve, Half-Sine curve and a Quintic Curve, different program function files are written using MATLAB and called in the general program code for

execution which give the outputs for the desired motion, velocity variation and acceleration variation w.r.t. time. These plots are obtained for each curve in both X and Z direction of motion and overall plots of X vs Z direction which are analysed for comparison.

TRAJECTORY GENERATION USING CUBIC FUNCTIONS

Theoretical Calculations in X direction:

Point 1-2 Starting Phase:

Motion followed is that of constant velocity, graphically it would be seen as a straight line with equation $S = S_{\text{initial}} + Vt$

$$T_{\text{case}} = [0, T_r/2] = [0, T/4]$$

$$T_{\text{overall}} = [0, T_r/2] = [0, T/4]$$

Here $S = -S_0/2$ and $t = T_r/2 = T/4$

$$\text{Hence } V = (-S_0/2) / (T/4) = -2S_0/T$$

Thus equation to be plotted is $X(t) = (S_0/2) - (2S_0/T)t$

Point 2-3-4 Swing Phase:

During the swing phase a cubic function is used to describe the motion.

General form of the cubic curve is

$$X(t) = C_0 + C_1t + C_2t^2 + C_3t^3$$

$$T_{\text{case}} = [0, T_y] = [0, T/2]$$

$$T_{\text{overall}} = [T_r/2, T_r/2 + T_y] = [T/4, 3T/4]$$

We know the initial and final positions as well as velocities thus we calculated the coefficients by solving the system of equations and obtained

$$C_0 = 0$$

$$C_1 = (-2S_0)/T$$

$$C_2 = (24 S_0)/T^2$$

$$C_3 = (-32 S_0)/T^3$$

Thus the equation of motion becomes

$$X(t) = ((-2S_0)/T)t + ((24 S_0)/T^2)t^2 + ((-32 S_0)/T^3)t^3$$

$$\dot{X}(t) = ((-2S_0)/T) + ((48 S_0)/T^2)t + ((-96 S_0)/T^3)t^2$$

$$\ddot{X}(t) = ((48 S_0)/T^2) + ((-192S_0)/T^3)t$$

Point 4-1-2 Stance phase

Motion followed is that of constant velocity, graphically it would be seen as a straight line with equation $S = S_{\text{initial}} + Vt$

$$T_{\text{case}} = [0, T_r] = [0, T/2]$$

$$T_{\text{overall}} = [T_y + T_r/2, T_y + 3T_r/2] = [3T/4, 5T/4]$$

Here $S = -S_0$ and $t = T_r = T/2$

$$\text{Hence } V = (-S_0)/(T/2) = -2S_0/T$$

Thus equation to be plotted is $X(t) = (S_0) - (2S_0/T)t$

Output Graphs:

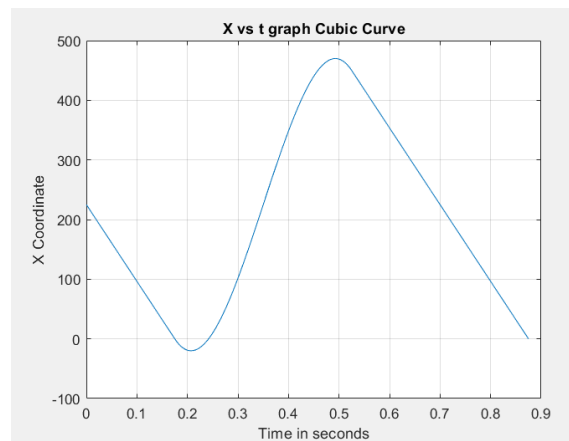


Figure: X vs T graph for Cubic trajectory

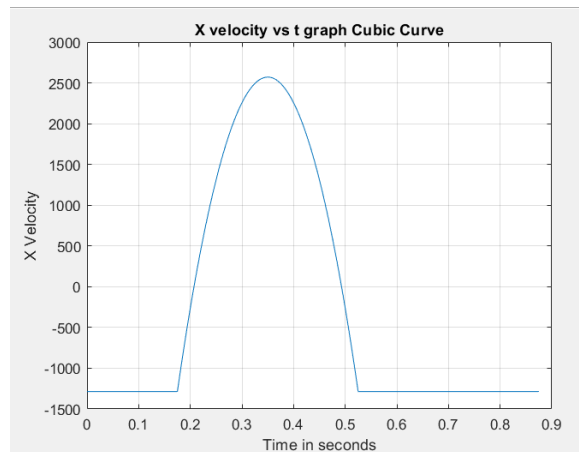


Figure: X-velocity vs T graph for Cubic trajectory

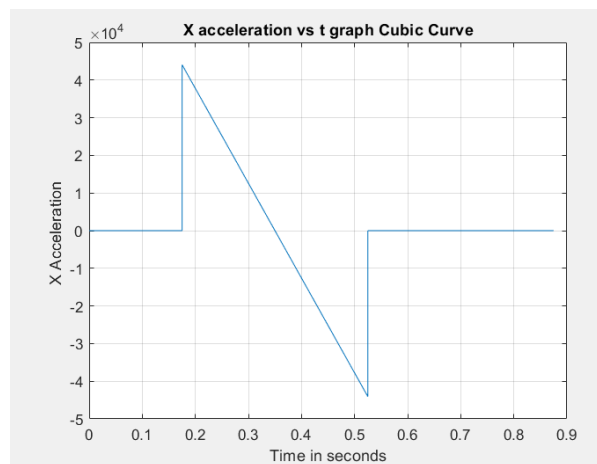


Figure: X-acceleration vs T graph for Cubic trajectory

Theoretical Calculations in Z direction:

Point 1-2 Starting Phase:

There is no motion during this phase.

$$T_{\text{case}} = [0, T_r/2] = [0, T/4]$$

$$T_{\text{overall}} = [0, T_r/2] = [0, T/4]$$

Point 2-3 Rise:

The initial and final velocity are zero and height attained is H_0 , using a cubic form to describe the motion

General form of the cubic curve is

$$Z(t) = C_0 + C_1t + C_2t^2 + C_3t^3$$

Using the boundary conditions we get values of coefficients as

$$C_0 = 0$$

$$C_1 = 0$$

$$C_2 = (48H_0)/T^2$$

$$C_3 = (-128H_0)/T^3$$

And equation of motion becomes

$$Z(t) = (48H_0/T^2)t^2 + ((-128H_0)/T^3)t^3$$

$$\dot{Z}(t) = (96H_0/T^2)t + ((-384H_0)/T^3)t^2$$

$$\ddot{Z}(t) = (96H_0/T^2) + ((-768H_0)/T^3)t$$

Point 3-4 Fall:

The initial and final velocity are zero and initial height is H_0 and final height is zero , using a cubic form to describe the motion

General form of the cubic curve is

$$Z(t) = C_0 + C_1t + C_2t^2 + C_3t^3$$

Using the boundary conditions we get values of coefficients as

$$C_0 = H_0$$

$$C_1 = 0$$

$$C_2 = (48H_0)/T^2$$

$$C_3 = (-128H_0)/T^3$$

And equation of motion becomes

$$Z(t) = H_0 + ((-48H_0)/T^2)t^2 + ((128H_0)/T^3)t^3$$

$$\dot{Z}(t) = ((-96H_0)/T^2)t + ((384H_0)/T^3)t^2$$

$$\ddot{Z}(t) = ((-96H_0)/T^2) + ((768H_0)/T^3)t$$

Point 4-1-2 Stance Phase:

There is no motion during this phase.

Output Graphs:

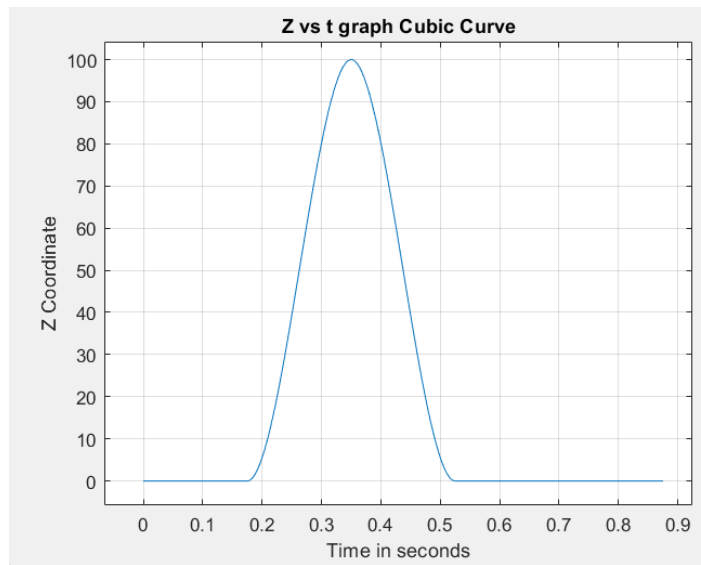


Figure: Z vs T graph for Cubic trajectory

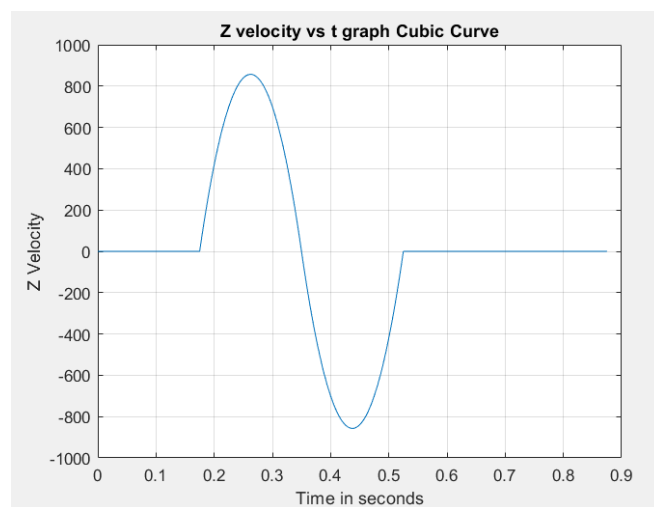


Figure: Z-velocity vs T graph for Cubic trajectory

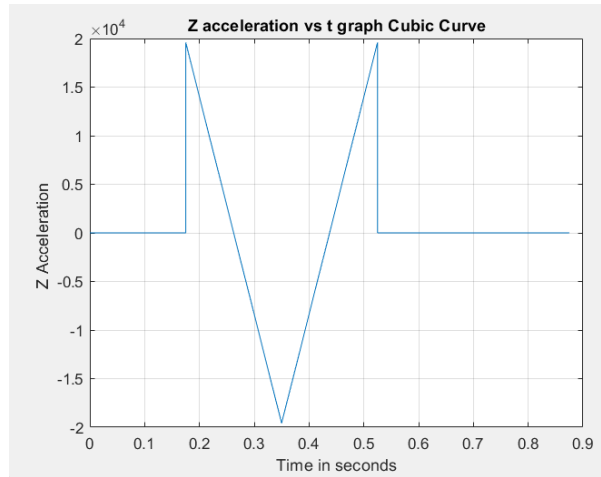


Figure: Z-acceleration vs T graph for Cubic trajectory

Overall X vs Z motion output graph

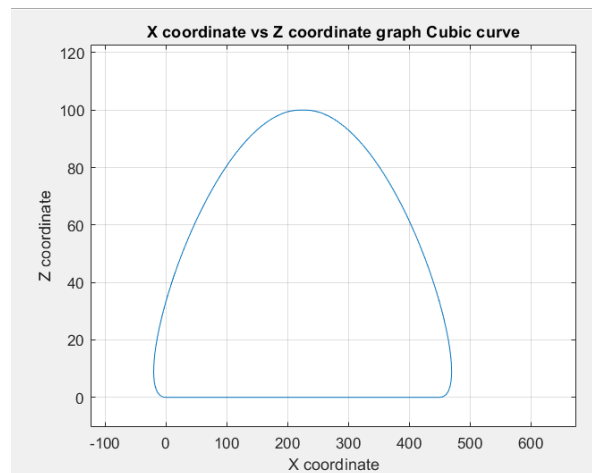


Figure: Overall X-Z motion for Cubic trajectory

TRAJECTORY GENERATION USING HALF-SINE FUNCTIONS

We will use multiple half-sine functions to generate overall trajectory, since this trajectory is a piecewise function of half-sine functions divided over different parts of the walking cycle the trajectory is thus also referred to as composite cycloid curve.

Theoretical Calculations in X direction:

Point 1-2 Starting Phase:

$$T_{\text{case}} = [0, T_r/2] = [0, T/4]$$

$$T_{\text{overall}} = [0, T_r/2] = [0, T/4]$$

Equation for Acceleration from $T=0$ to $T=T_r/2$:

$$\ddot{x}(t) = 2k \cdot \sin(4\pi \cdot t_n)$$

Equation for Velocity from $T=0$ to $T= T_r/2$:

$$\dot{x}(t) = \frac{k \cdot T_y}{2\pi} \cdot (1 - \cos(4\pi \cdot t_n))$$

Equation for X position $T=0$ to $T= T_r/2$:

$$x(t) = \frac{k \cdot T_y^2}{2\pi} \cdot \left(t_n - \frac{\sin(4\pi \cdot t_n)}{4\pi} \right) + 0.5(S_0)$$

Where constant 'k' is :

$$k = \frac{-2\pi \cdot S_0}{T_y^2}$$

Where t_n is:

$$t_n = \frac{t}{T_y}$$

Point 2-3-4 Swing Phase:

$$T_{\text{case}} = [0, T_y] = [0, T/2]$$

$$T_{\text{overall}} = [T_r/2, T_r/2 + T_y] = [T/4, 3T/4]$$

Equation for Acceleration from $T=0$ to $T=T_y$:

$$\ddot{x}(t) = k \cdot \sin(2\pi \cdot t_n)$$

Equation for Velocity from T=0 to T=T_y :

$$\dot{x}(t) = \frac{k \cdot T_y}{2\pi} \cdot (1 - \cos(2\pi \cdot t_n))$$

Equation for X position T=0 to T=T_y :

$$x(t) = \frac{k \cdot T_y^2}{2\pi} \cdot \left(t_n - \frac{\sin(2\pi \cdot t_n)}{2\pi} \right)$$

Where constant 'k' is :

$$k = \frac{2\pi \cdot S_0}{T_y^2}$$

Where t_n is:

$$t_n = \frac{t}{T_y}$$

Point 4-1-2 Stance phase

$$T_{\text{case}} = [0, T_r] = [0, T/2]$$

$$T_{\text{overall}} = [T_y + T_r/2, T_y + 3T_r/2] = [3T/4, 5T/4]$$

Equation for Acceleration from T=0 to T=T_r :

$$\ddot{x}(t) = k \cdot \sin(2\pi \cdot t_n)$$

Equation for Velocity from T=0 to T=T_r :

$$\dot{x}(t) = \frac{k \cdot T_y}{2\pi} \cdot (1 - \cos(2\pi \cdot t_n))$$

Equation for X position T=0 to T=T_r :

$$x(t) = \frac{k \cdot T_y^2}{2\pi} \cdot \left(t_n - \frac{\sin(2\pi \cdot t_n)}{2\pi} \right) + S_0$$

Where constant 'k' is :

$$k = \frac{-2\pi.S_0}{T_y^2}$$

Where t_n is:

$$t_n = \frac{t}{T_y}$$

Output Graphs:

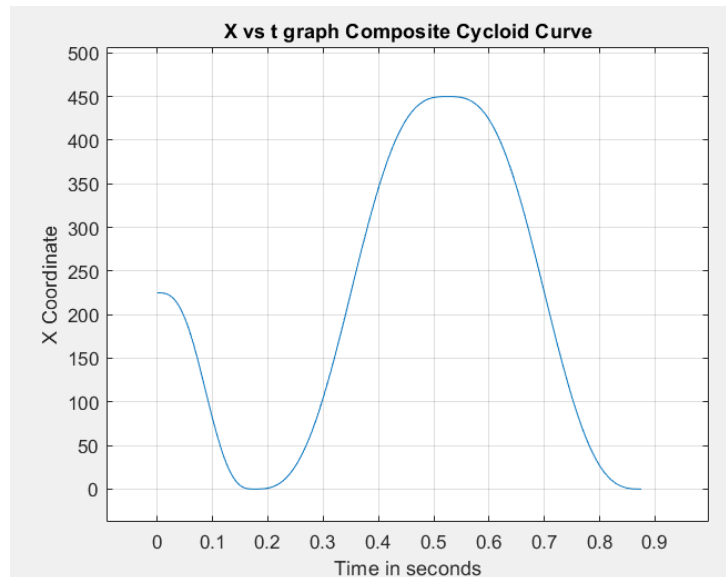


Figure: X vs T graph for composite cycloid trajectory

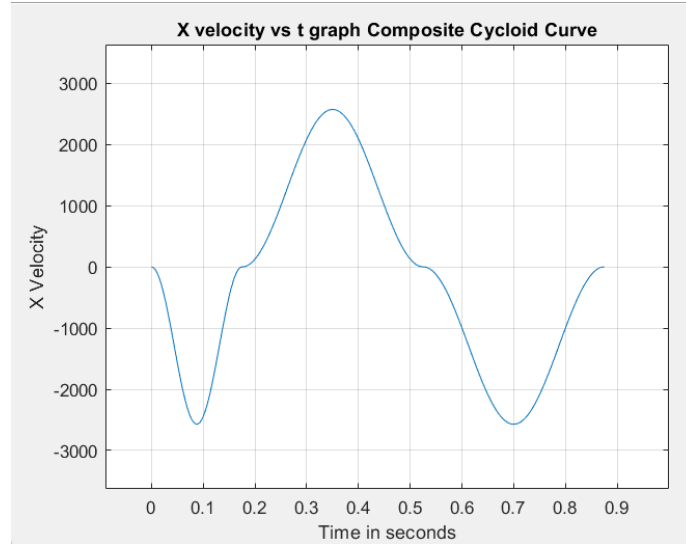


Figure: X-velocity vs T graph for composite cycloid trajectory

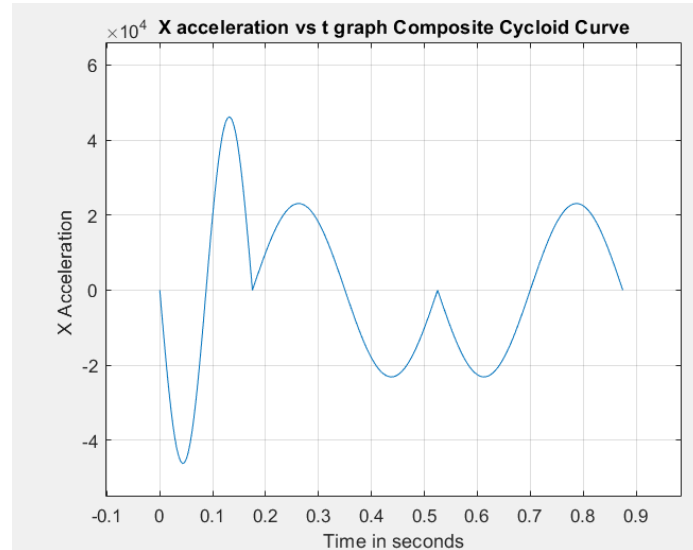


Figure: X-acceleration vs T graph for composite cycloid trajectory

Theoretical Calculations in Z direction:

Point 1-2 Starting Phase:

There is no motion during this phase.

$$T_{\text{case}} = [0, T_r/2] = [0, T/4]$$

$$T_{\text{overall}} = [0, T_r/2] = [0, T/4]$$

Point 2-3 Rise:

The initial and final velocity are zero and height attained is H_0

Equation for acceleration:

$$\ddot{z}(t) = k \cdot \sin\left(4\pi \frac{t}{T_y}\right)$$

Equation for Velocity:

$$\dot{z}(t) = \frac{kT_y}{4\pi} \cdot \left(1 - \cos\left(4\pi \cdot \frac{t}{T_y}\right)\right)$$

Equation for position:

$$z(t) = \frac{kT_y}{4\pi} \left(\frac{T_y}{2} - \frac{T_y}{4\pi} \sin\left(4\pi \frac{t}{T_y}\right)\right)$$

Where k is :

$$k = \frac{8\pi H_0}{T_y^2}$$

Point 3-4 Fall:

The initial and final velocity are zero and initial height is H_0 and final height is zero

Equation for acceleration:

$$\ddot{z}(t) = -k \cdot \sin\left(4\pi \frac{t}{T_y}\right)$$

Equation for Velocity:

$$\dot{z}(t) = \frac{-kT_y}{4\pi} \cdot \left(1 - \cos\left(4\pi \cdot \frac{t}{T_y}\right)\right)$$

Equation for position:

$$z(t) = H - \frac{kT_y}{4\pi} \left(\frac{T_y}{2} - \frac{T_y}{4\pi} \sin\left(4\pi \frac{t}{T_y}\right)\right)$$

Where k is :

$$k = \frac{8\pi H_0}{T_y^2}$$

Point 4-1-2 Stance Phase:

There is no motion during this phase.

Output Graphs:

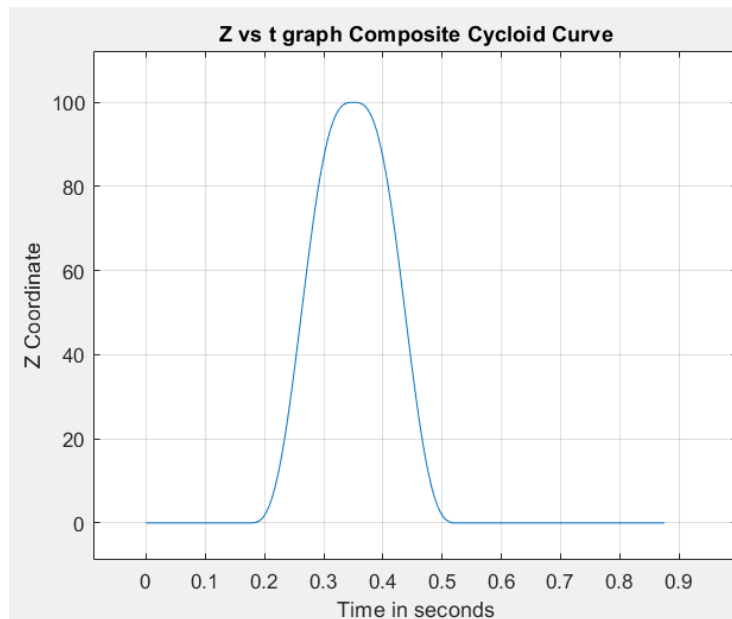


Figure: Z vs T graph for composite cycloid trajectory

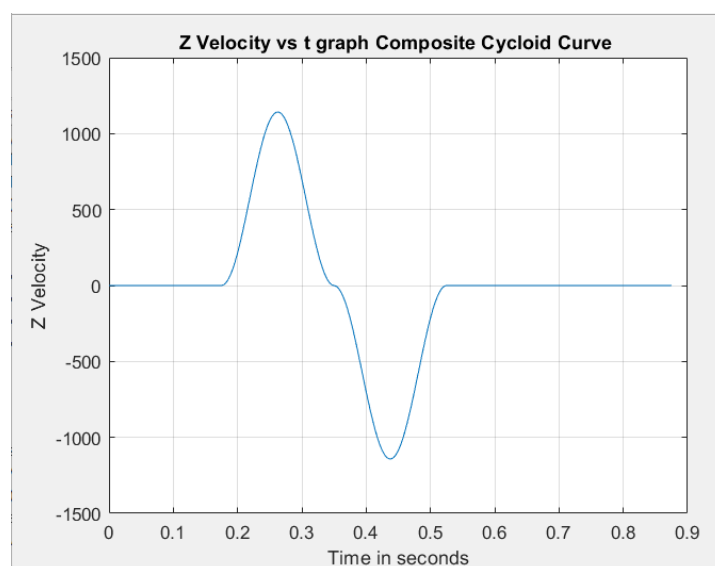


Figure: Z-velocity vs T graph for composite cycloid trajectory

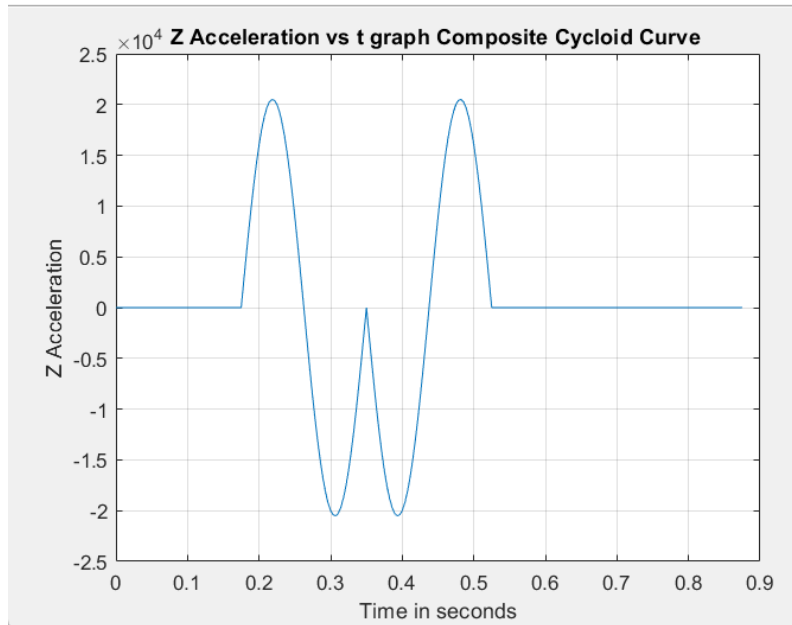


Figure: Z-acceleration vs T graph for composite cycloid trajectory

Overall X vs Z motion output graph:

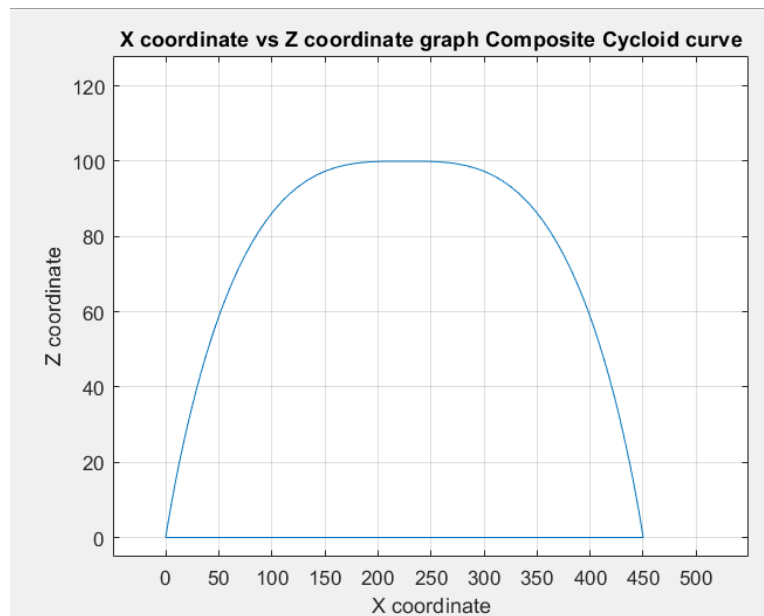


Figure: Overall X-Z motion for composite cycloid trajectory

TRAJECTORY GENERATION USING QUINTIC FUNCTIONS

Theoretical Calculations in X direction

Point 1-2 Starting Phase:

Motion followed is that of constant velocity, graphically it would be seen as a straight line with equation $S=S_{\text{initial}} + Vt$

$$T_{\text{case}} = [0, T_r/2]=[0, T/4]$$

$$T_{\text{overall}} = [0, T_r/2] = [0, T/4]$$

Here $S=-S_0/2$ and $t=T_r/2=T/4$

$$\text{Hence } V=(-S_0/2)/(T/4) = -2S_0/T$$

Thus equation to be plotted is $X(t)=(S_0/2)-(2S_0/T)t$

Point 2-3 Rise:

The final velocity is zero and distance covered is $S_0/2$, using a quintic form to describe the motion.

General form of the Quintic curve is

$$X(t) = C_0 + C_1t + C_2t^2 + C_3t^3 + C_4t^4 + C_5t^5$$

Using the boundary conditions we get values of coefficients as

$$C_0 = 0$$

$$C_1 = -2S_0/T$$

$$C_2 = 0$$

$$C_3 = 512S_0/T^3$$

$$C_4 = -2944S_0/T^4$$

$$C_5 = 4608S_0/T^5$$

And equation of motion becomes

$$X(t) = (-2S_0/T)t + (512S_0/T^3)t^3 + (-2944S_0/T^4)t^4 + (4608S_0/T^5)t^5$$

$$\dot{X}(t) = (-2S_0/T) + 3(512S_0/T^3)t^2 + 4(-2944S_0/T^4)t^3 + 5(4608S_0/T^5)t^4$$

$$\ddot{X}(t) = 6(512S_0/T^3)t + 12(-2944S_0/T^4)t^2 + 20(4608S_0/T^5)t^3$$

Point 3-4 Fall:

The initial velocity is zero and initial position is $S_0/2$ and final position is S_0 , using a quintic curve to describe the motion

General form of the Quintic curve is

$$X(t) = C_0 + C_1t + C_2t^2 + C_3t^3 + C_4t^4 + C_5t^5$$

Using the boundary conditions we get values of coefficients as

$$C_0 = S_0/2$$

$$C_1 = 0$$

$$C_2 = 0$$

$$C_3 = 448S_0/T^3$$

$$C_4 = -2816S_0/T^4$$

$$C_5 = 4608S_0/T^5$$

And equation of motion becomes

$$X(t) = (S_0/2) + (448S_0/T^3)t^3 + (-2816S_0/T^4)t^4 + (4608S_0/T^5)t^5$$

$$\dot{X}(t) = 3(448S_0/T^3)t^2 + 4(-2816S_0/T^4)t^3 + 5(4608S_0/T^5)t^4$$

$$\ddot{X}(t) = 6(448S_0/T^3)t + 12(-2816S_0/T^4)t^2 + 20(4608S_0/T^5)t^3$$

Point 4-1-2 Stance phase

Motion followed is that of constant velocity, graphically it would be seen as a straight line with equation $S=S_{\text{initial}} + Vt$

$$T_{\text{case}} = [0, T_r]=[0, T/2]$$

$$T_{\text{overall}} = [T_y + T_r/2, T_y + 3T_r/2] = [3T/4, 5T/4]$$

Here $S=-S_0$ and $t=T_r=T/2$

$$\text{Hence } V=(-S_0)/(T/2) = -2S_0/T$$

Thus equation to be plotted is $X(t)=(S_0)-(2S_0/T)t$

Output Graphs

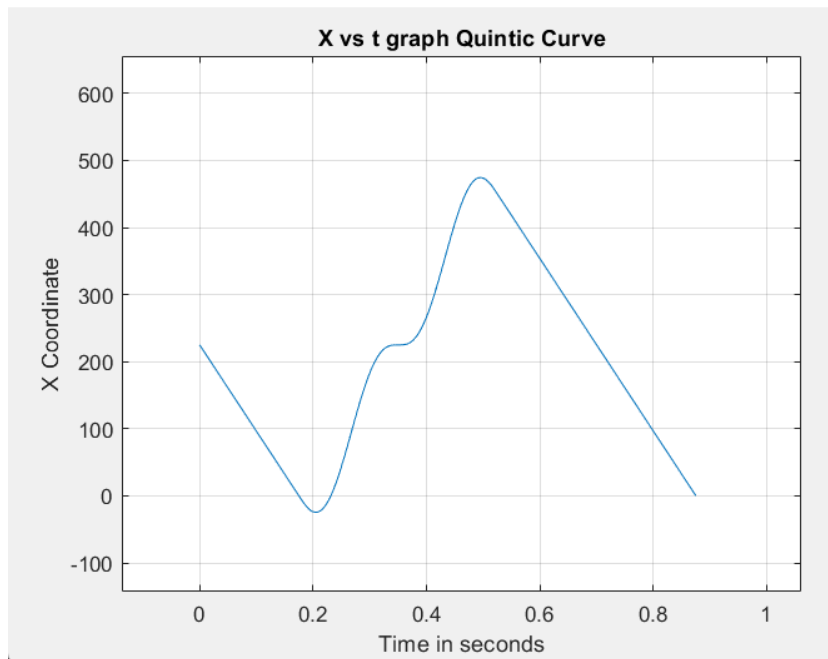


Figure: X vs T graph for Quintic trajectory

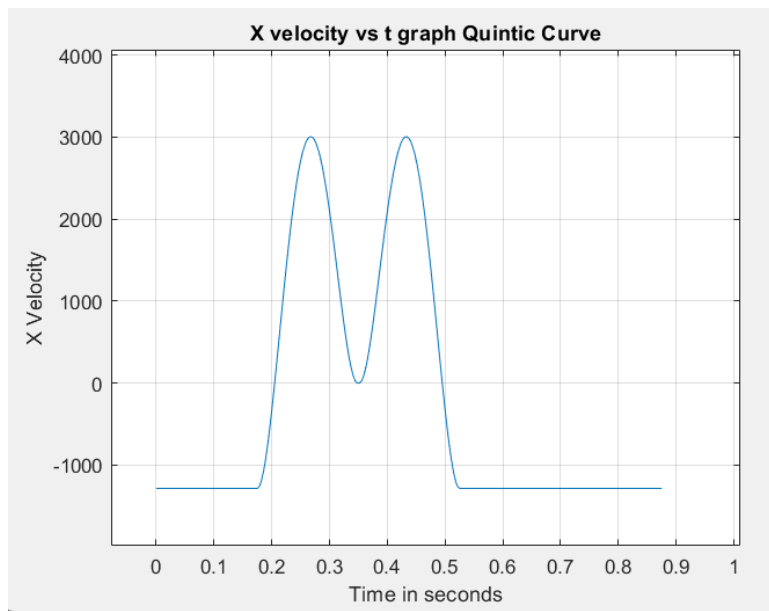


Figure: X-velocity vs T graph for Quintic trajectory

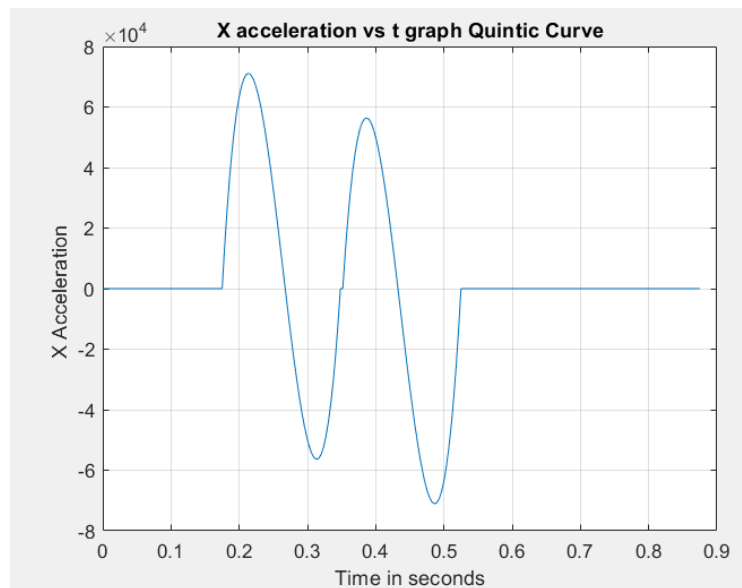


Figure: X-acceleration vs T graph for Quintic trajectory

Theoretical Calculations in Z direction

Point 1-2 Starting Phase:

There is no motion during this phase.

$$T_{\text{case}} = [0, T_r/2] = [0, T/4]$$

$$T_{\text{overall}} = [0, T_r/2] = [0, T/4]$$

Point 2-3 Rise:

The initial and final velocity are zero and height attained is H_0 , using a quintic curve to describe the motion

General form of the quintic curve is

$$Z(t) = C_0 + C_1t + C_2t^2 + C_3t^3 + C_4t^4 + C_5t^5$$

Using the boundary conditions we get values of coefficients as

$$C_0 = 0$$

$$C_1 = 0$$

$$C_2 = 0$$

$$C_3 = 640H_0/T^3$$

$$C_4 = -3840H_0/T^4$$

$$C_5 = 6144H_0/T^5$$

And equation of motion becomes

$$Z(t) = (640H_0/T^3)t^3 + (-3840H_0/T^4)t^4 + (6144H_0/T^5)t^5$$

$$\dot{Z}(t) = 3(640H_0/T^3)t^2 + 4(-3840H_0/T^4)t^3 + 5(6144H_0/T^5)t^4$$

$$\ddot{Z}(t) = 6(640H_0/T^3)t + 12(-3840H_0/T^4)t^2 + 20(6144H_0/T^5)t^3$$

Point 3-4 Fall:

The initial and final velocity are zero and initial height is H_0 and final height is zero , using a quintic curve to describe the motion

General form of the quintic curve is

$$Z(t) = C_0 + C_1t + C_2t^2 + C_3t^3 + C_4t^4 + C_5t^5$$

Using the boundary conditions we get values of coefficients as

$$C_0 = H_0$$

$$C_1 = 0$$

$$C_2 = 0$$

$$C_3 = -640H_0/T^3$$

$$C_4 = 3840H_0/T^4$$

$$C_5 = -6144H_0/T^5$$

And equation of motion becomes

$$Z(t) = (H_0) + (-640H_0/T^3)t^3 + (3840H_0/T^4)t^4 + (-6144H_0/T^5)t^5$$

$$\dot{Z}(t) = 3(-640H_0/T^3)t^2 + 4(3840H_0/T^4)t^3 + 5(-6144H_0/T^5)t^4$$

$$\ddot{Z}(t) = 6(-640H_0/T^3)t + 12(3840H_0/T^4)t^2 + 20(-6144H_0/T^5)t^3$$

Point 4-1-2 Stance Phase:

There is no motion during this phase.

Output Graphs

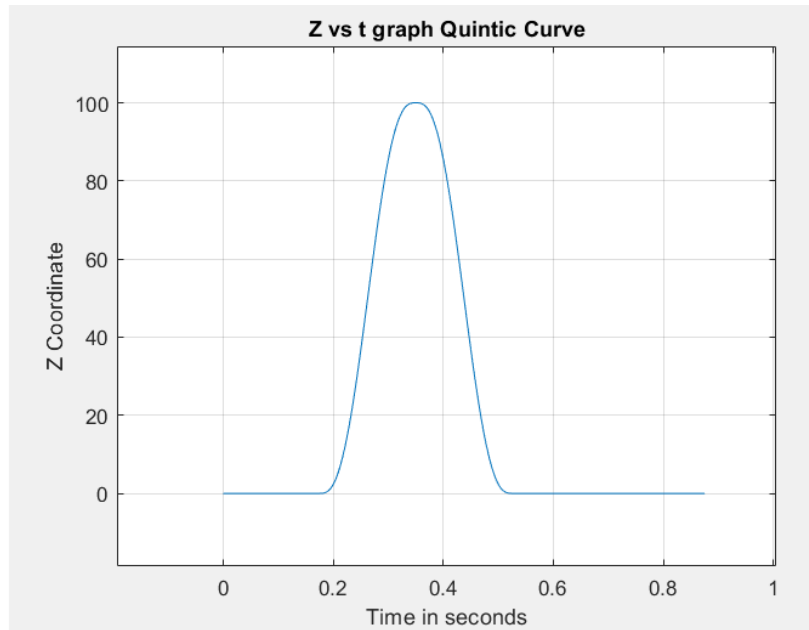


Figure: Z vs T graph for Quintic trajectory

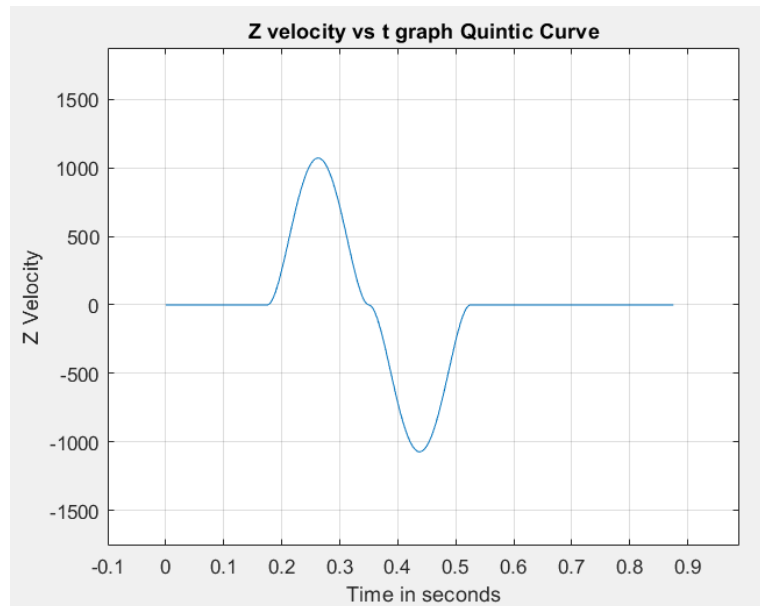


Figure: Z-velocity vs T graph for Quintic trajectory

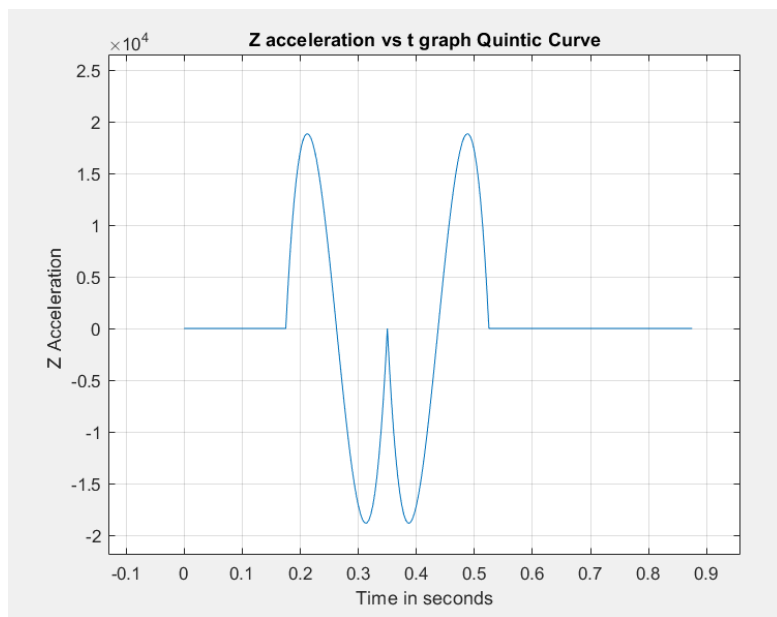


Figure: Z-acceleration vs T graph for Quintic trajectory

Overall X vs Z motion output graph

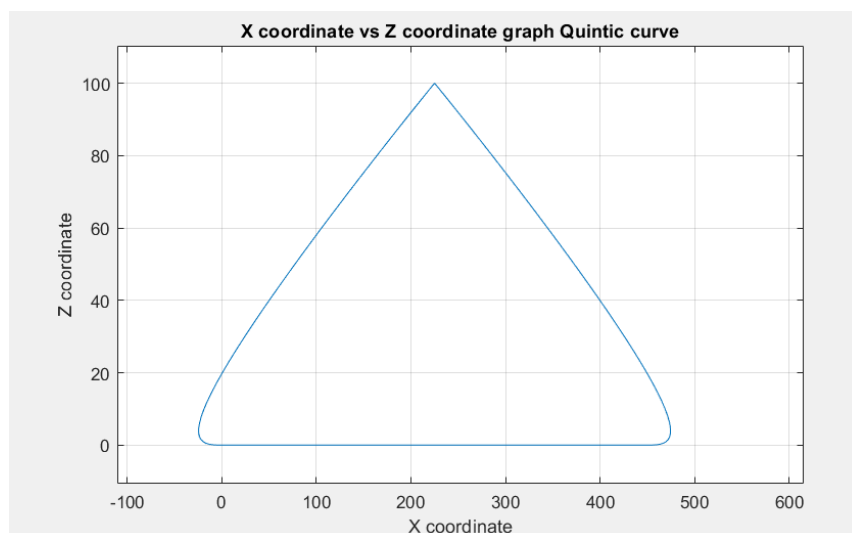


Figure: Overall X-Z Motion for Quintic trajectory